RecoverLens

MOVE BETTER · RECOVER SMARTER

FIFTH ATU AFRICA INNOVATION CHALLENGE · PROPOSAL

RecoverLens: an AI rehabilitation mirror for every home

Turning any phone with a camera into a physiotherapy coach that counts every rep, corrects form by voice in the patient's own language, and keeps the physiotherapist in the loop.

Lead innovator	Harry James Makoni
Ecosystem	Code for Better
Host institution	National University of Science and Technology (NUST), Bulawayo, Zimbabwe
AI field	Computer vision: on-device pose estimation for health
Live app	recover-lens.vercel.app
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1. Executive summary

RecoverLens turns any phone, tablet or laptop with a camera into an AI rehabilitation mirror. It counts every repetition, checks form as the patient moves and coaches by voice in the patient's own language, starting with English, chiShona and isiNdebele, with more African languages to follow. Physiotherapists follow every session remotely. It was built at NUST within the Code for Better ecosystem, and it is live today.

The problem. Millions of Africans need rehabilitation after surgery, injury or stroke, but physiotherapists are few and far away. Most patients go home with a printed exercise sheet. Nobody knows whether they do the exercises, or do them correctly, until the next visit.

The solution. The AI tracks 33 points on the body with on-device pose estimation. It turns them into joint angles, repetitions, a form score and spoken corrections in real time. It also spots signs of tiring, compares each session with the patient's own baseline, estimates the load on the joint and projects when the clinician's target will be reached. Video never leaves the device, and the app works offline after the first visit.

Where it stands. The platform is a working, public web app. It offers 13 rehabilitation exercises, 4 sports injury screening tests and 4 user roles (patient, caregiver, physiotherapist, coach). It is covered by 157 automated tests and has been checked against the WCAG 2.2 AA accessibility standard.

The ask. Recognition and support from ATU will fund a 12-month supervised pilot with partner physiotherapy practices and schools. It will also fund clinical tuning of the AI, test devices and cloud hosting, and certification for learners in the Code for Better ecosystem.

2.4 billion
people worldwide could benefit from rehabilitation

1 per 37,000
practising physiotherapists to people in Zimbabwe

Up to 70%
of patients do not keep up with physiotherapy

Sources: PAHO/WHO; World Physiotherapy (2024); Jack et al. (2010).

2. The problem

Across Africa, rehabilitation is out of reach for most people who need it. In Zimbabwe, where RecoverLens was built, there is about one practising physiotherapist for every 37,000 people, and rural patients who need rehabilitation are sent home with a printed sheet and no way to check their progress.

The need is large. About 2.4 billion people worldwide live with a health condition that could benefit from rehabilitation (PAHO/WHO). In some low- and middle-income countries, more than half of the people who need rehabilitation do not receive it.

Practitioners are scarce. Many low- and middle-income countries have fewer than 10 skilled rehabilitation practitioners per million people (WHO). Zimbabwe reported 442 practising physiotherapists in 2024, or 0.27 per 10,000 people (World Physiotherapy).

Distance and cost stop people attending. A patient in Binga or Gokwe recovering from knee surgery may need several bus journeys to reach a physiotherapist. Each visit costs fares, a day's earnings and the physiotherapist's time.

Home exercise fails without feedback. Non-adherence to physiotherapy can be as high as 70%, and is worst in unsupervised home programmes (Jack et al., 2010). A printed sheet cannot tell a patient that a knee is caving in, that an arm is not lifting high enough or that they are tiring. The physiotherapist does not know whether the exercises were done at all until the next visit, weeks later.

The same gap exists in sport. Screening athletes for movement patterns linked to injury usually needs a sports physiotherapist. Most school and club teams do not have one.

The result is slower recovery, avoidable re-injury and patients lost to follow-up. Physiotherapists cannot see, between visits, which patients need them most.

3. The solution: RecoverLens

RecoverLens is a rehabilitation coach that sees. The patient props up a phone, steps back and exercises. The app watches through the camera, counts each repetition, judges its form and speaks a correction in the patient's language. The physiotherapist sees the result the same day.

It runs in the web browser of any phone, tablet or laptop with a camera. There is nothing to install, and after the first visit it works without a connection.

3.1 The patient's session

1. The physiotherapist assigns exercises, sets and a target angle for each patient.
2. The patient opens today's programme and answers a short pain check-in.
3. The app checks that the whole body is in view, then counts down.
4. During each set, the screen acts as a mirror with the skeleton drawn on the body. The voice coach counts reps and corrects form: "Raise both arms evenly."
5. If reps slow down or shorten, the coach suggests a rest. It never stops the patient exercising.
6. A summary shows correct reps, range of movement, form score, signs of tiring and joint load. It compares them with the patient's own recent sessions.
7. The session is saved and synced to the physiotherapist. The patient can share it with family on WhatsApp.

3.2 Who uses it

User	What RecoverLens does for them
Patient	Coached home exercise with live feedback, in their own language

User	What RecoverLens does for them
Caregiver	A mode that coaches the family member or helper who supports the patient at home
Physiotherapist	A portal with every patient's sessions, trends and alerts when pain rises, scores drop or sessions are missed; reports to print or export
Sports coach	Squat, single-leg squat, balance and landing screens that flag movement patterns, such as knees caving inward, to work on early

3.3 What patients can do today

- **13 rehabilitation exercises** for the shoulder, elbow and knee. Six are measured live by the AI: shoulder abduction, shoulder flexion, elbow curl, mini squat, sit-to-stand and single-leg balance. Seven are guided by voice and timer.
- **4 sports screening tests** for school and club athletes.
- **African languages first:** every screen and voice cue is in English, chiShona and isiNdebele today, the first languages of the pilot. More African languages are added as the platform reaches new countries.
- **Accessibility built in:** large and extra-large text, high contrast and full voice guidance, so people with low vision can follow a session.

4. How the AI works

RecoverLens pairs a proven pose-estimation model with an exercise engine built at NUST. The engine turns body points into clinical measures and spoken coaching, and all of it runs on the patient's own device.

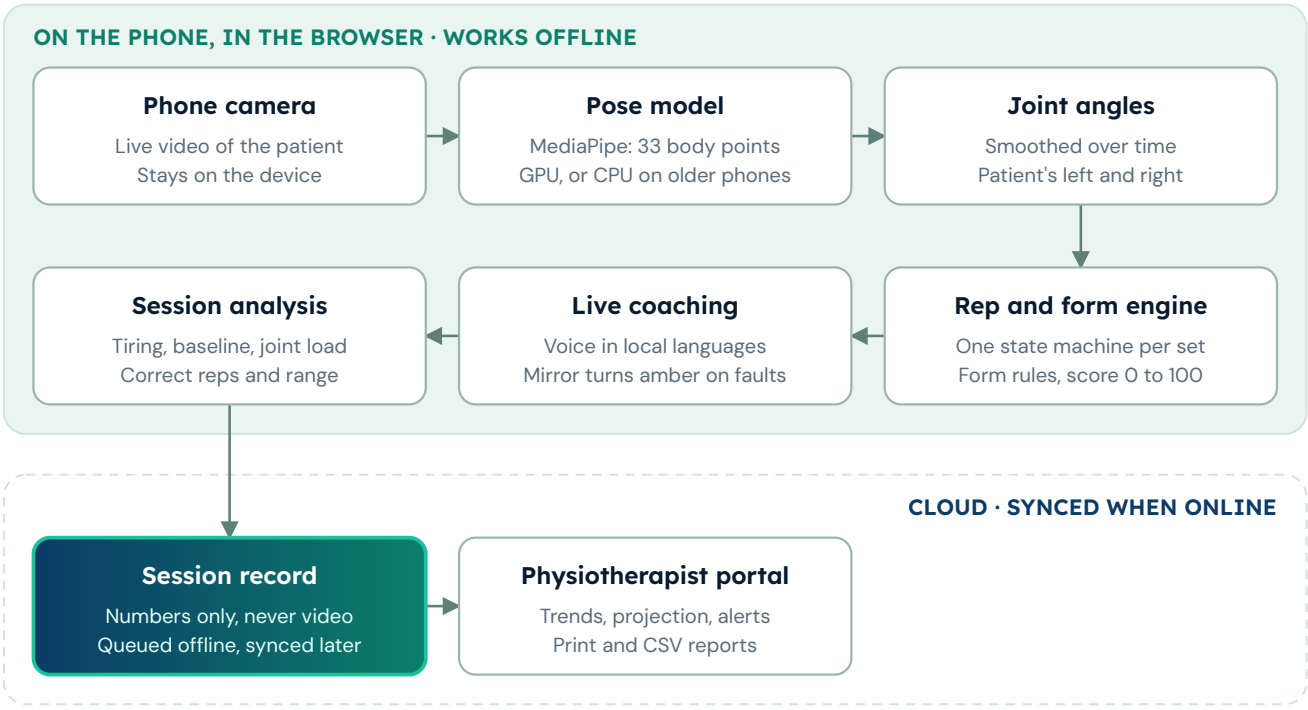


Figure 1. All the AI runs on the phone; only session numbers reach the physiotherapist.

4.1 From camera to coaching

1. **Seeing the body.** Google's MediaPipe Pose Landmarker finds 33 body points in each video frame, inside the browser. It uses the phone's graphics chip and falls back to the processor on phones that cannot.
2. **Measuring.** The engine computes shoulder, elbow, hip and knee angles and smooths them over time, so a slow phone does not miss reps. Front-view squats use 3D points, because a knee bending towards the camera barely changes its flat angle. Depth is measured from each person's own standing angle.
3. **Counting and judging.** One state machine per set counts every completed rep and tracks correct reps separately. Form rules detect leaning, uneven arms, a bent elbow, an arched back, knees caving inward and weight shifting. Each rep gets a movement score from 0 to 100.
4. **Coaching.** A voice queue speaks counts and corrections in the patient's language (English, chiShona and isiNdebele today), with safety cues first. The mirror turns amber on a fault, and the on-screen panel never covers the patient's body.

4.2 The AI coach: beyond counting

Capability	What it measures	Why it matters
Signs of tiring	The last third of a session's reps against the first third: 15% slower, 8% less range or 8 points lower form	Tells the patient to rest before form breaks down; shows the physiotherapist who is overloaded
Live pacing	Within a set, the latest 3 reps against the first 3	The coach suggests a rest once per set, without stopping the exercise
Personal baseline	Each session against the patient's own last 3 sessions of that exercise	Progress is judged against the person, not a population average

Capability	What it measures	Why it matters
Joint load	The turning force on the elbow or shoulder, in newton-metres, from limb mass and anything held	Lets the physiotherapist add weight safely; checked against the OpenSim research model, $R^2 = 0.85$
Recovery projection	The trend in range of movement across sessions	Estimates how many sessions remain until the clinician's target angle
Clinician insights	Score and range trends, exercise days, recurring faults, pain after exercise	Alerts the physiotherapist to the patients who need a call

> All results are worded as observations for a clinician to review, never as a diagnosis. Thresholds that need real-world tuning are marked in the code and will be set during the pilot.

4.3 Why on the device

- **Privacy:** video is processed in memory and never uploaded or stored.
- **Low bandwidth:** after the first visit, a session needs no data at all.
- **Low cost at scale:** each new user brings their own processor, so server costs barely grow.

5. Innovation: what is new

RecoverLens brings together what current options offer only separately: live feedback on every rep, no travel, local languages and a physiotherapist who sees every home session.

Option	Feedback while exercising	Travel or data needed	Local languages	Physiotherapist sees home sessions
Printed exercise sheet	None	None	Rarely	No
Clinic visit	Yes, from the physiotherapist	Travel, fares and time off work	Yes	Only in the clinic
Video-call telerehabilitation	Yes, while the call lasts	A steady connection and the physiotherapist's time	Depends on the physiotherapist	Only during the call
Exercise video apps	None; the patient copies a video	Data to stream video	Usually English only	No
RecoverLens	**Every rep, by voice, in real time**	**None after the first visit**	**English, chiShona, isiNdebele today; more to follow**	**Every session, with alerts**

5.1 What makes it distinctive

- **It measures, not just instructs.** A phone camera yields joint angles, correct reps, form faults, signs of tiring and joint load. These are measures a physiotherapist would otherwise have to see in person.
- **It is personal.** Progress is judged against the patient's own recent sessions, and the projection counts down to the target the physiotherapist set for that person.
- **It is built for African conditions.** It works offline, runs on the processor of older phones and keeps counting at 4 frames per second. It speaks the patient's language, starting with chiShona and isiNdebele, coaches caregivers and shares summaries on WhatsApp.
- **It is private by design.** Video never leaves the device, which removes the largest privacy risk of camera-based health tools.
- **One engine serves two markets.** The same AI powers home rehabilitation for patients and injury screening for school and club athletes.

- **It is inclusive.** It is built to the WCAG 2.2 AA accessibility standard, with voice guidance throughout for people with low vision.

6. Relevance to Africa's digital priorities

RecoverLens applies AI to digital health, one of the critical sectors of Africa's digital transformation. It does so in a way that suits African networks, devices and languages.

Priority	How RecoverLens contributes
AU Digital Transformation Strategy for Africa 2020–2030: digital health and digital inclusion	Brings AI-guided care to any phone with a camera, offline, in local languages
UN Sustainable Development Goal 3.8: universal health coverage	Extends rehabilitation to patients who cannot travel to a clinic
WHO Rehabilitation 2030 and World Health Assembly resolution WHA76.6 on strengthening rehabilitation	Multiplies the reach of each physiotherapist through remote monitoring
SDG 10: reduced inequalities	Serves rural patients, people with low vision and speakers of African languages
Zimbabwe Vision 2030 and the National Development Strategy: health, digital economy and innovation	Home-grown AI from a Zimbabwean university, built for local health workers
Local content and data sovereignty	Video stays on the device; the software and data can be hosted in-country

Why it matters now. Far more homes have a smartphone than have a physiotherapist within reach. RecoverLens turns that phone into a link between the patient at home and the clinician in town.

7. Stage of development and technical readiness

RecoverLens is a working, publicly deployed prototype at about Technology Readiness Level 5: validated on recorded and synthetic movement, not yet with patients. The pilot will take it to Level 7, proven with real users in clinics and homes.

7.1 Evidence of development and testing

Evidence	Result
Live deployment	Public web app with a Python API, at recover-lens.vercel.app
Automated tests	135 frontend tests (pose engine, voice, screening, translations) and 22 backend API tests, all passing
Engine tests on a synthetic body	10 of 10 reps counted with pauses; a lean flagged within 0.5 s; counting holds at 4 frames per second
Calibration on real pose data	Shoulder thresholds retuned on real MediaPipe output from reference photos of a person
Joint-load validation	$R^2 = 0.85$ against OpenSim elbow torque on a public 8-person dataset, leaving each person out in turn
Accessibility	Automated WCAG 2.2 AA checks pass on every screen, at phone and desktop widths, with extra-large text and high contrast
Languages	Every screen and voice cue in chiShona and isiNdebele; tests fail if any is missing
Validation tools	In-app landmark recorder, video-to-landmark extraction and replay evaluation, so real recordings become regression tests

7.2 Technology

- **Frontend:** React, TypeScript and Vite, installable as a progressive web app that works offline.
- **AI:** MediaPipe Pose Landmarker in the browser, plus a pure TypeScript engine for angles, reps, form, fatigue and joint load.
- **Backend:** Python FastAPI with SQLModel, on PostgreSQL or SQLite; the database upgrades itself when new fields are added.
- **Hosting:** Vercel today; portable to a Zimbabwean or university server.

7.3 Still to do before wider use

- Record 3 to 5 people and tune the movement thresholds, starting with squats.
- Native-speaker review of the chiShona and isiNdebele text.
- Test with a screen-reader user and a low-vision user.
- Test on iPhone Safari and Android Chrome in clinic lighting.
- Connect a persistent cloud database to the live site.
- Add user accounts with patient, caregiver, physiotherapist and coach roles.
- Clinical validation against a physiotherapist's assessment, with ethics approval.

8. The ecosystem: Code for Better at NUST

RecoverLens is fronted by Code for Better, the NUST innovation ecosystem through which students develop AI-based software into working products. Under Zimbabwe's Education 5.0 policy, innovation and industrialisation sit alongside teaching, research and community service as core university missions.

8.1 Host institution

The National University of Science and Technology (NUST) is a public university in Bulawayo, Zimbabwe, founded in 1991. It brings together computing, engineering and a medical school, which is the mix a health AI product needs.

8.2 Ecosystem profile

Element	Detail
Name	Code for Better
Host institution	National University of Science and Technology (NUST), Bulawayo, Zimbabwe
Purpose	[One sentence on Code for Better's mission]
Structure	[How Code for Better is organised and how it links to NUST faculties]
Active innovators	[Number of student teams supported in 2026]
Support mechanisms	[Mentorship, workspace, labs, peer learning]

8.3 Eligibility requirements and evidence

ATU requirement	Evidence from Code for Better
Institutional connection	RecoverLens is developed by a NUST student [programme, year] through Code for Better
Software focus	Code for Better supports AI-based software; RecoverLens is an AI computer-vision product
Named coordination	Code for Better's coordination structure guides and reviews member projects
Evidence of support	[Mentorship sessions, lab or device access, workspace, pitching practice received]
Learner nomination capacity	Code for Better can nominate learners for partner-supported training and certification (section 8.6)

8.4 Support RecoverLens has received

- Peer learning and code review within Code for Better.
- [Workspace, internet or devices provided]
- [Earlier pitching or competitions, such as TENT Gig 2026, and their outcome]

8.5 Partners and the support network

The pilot needs clinical and community partners. Letters of support will be attached where confirmed.

Partner type	Role in the project	Status
2 to 3 physiotherapy practices or hospital rehabilitation departments in Bulawayo	Pilot sites; prescribe exercises and review sessions	[Confirmed or in discussion]
A registered physiotherapist mentor	Checks each exercise, target and safety cue	[Name or in discussion]
NUST Faculty of Medicine	Clinical research supervision and ethics review	[To confirm]

Partner type	Role in the project	Status
2 schools or sports clubs	Pilot of the sports screening module	[Names or in discussion]
Native chiShona and isiNdebele speakers	Review of voice and screen text	[Names]

8.6 Learner pathway

Code for Better confirms it can nominate eligible learners or teams for any partner-supported training or certification package. RecoverLens will create places for at least 4 NUST students: 2 in software and AI, 1 in data and evaluation, and 1 from a health discipline. Their training areas would be cloud engineering, applied AI and machine learning, cybersecurity and data protection, and product management.

9. Deployment potential and sustainability

RecoverLens can scale cheaply, because each user's phone does the AI work and the app needs no store or install. Growth is driven by physiotherapists and coaches, who bring their patients and athletes with them.

9.1 Who pays and who benefits

Customer	What they pay	What they get
Physiotherapy practices and hospital departments	A monthly subscription [amount]	The clinician portal, alerts and reports; their patients use the app free
Schools and sports clubs	Per team, per term or season [amount]	Injury screening for their athletes and a coach portal
Individuals without a physiotherapist	About US$0.50 a week	Self-guided exercise programmes
NGOs, insurers and well-wishers	Sponsorship of patients	Rehabilitation for people who cannot pay, with progress reports

Patients never pay to use RecoverLens with their own physiotherapist.

9.2 Rollout

1. **Pilot, Bulawayo (2027):** 2 to 3 physiotherapy sites and 2 schools or clubs, supported by Code for Better.
2. **National (2027–2028):** Harare and provincial hospitals, through the physiotherapy professional bodies and NGO partners.
3. **Regional (from 2028):** other ATU Member States, starting with neighbours that share languages and referral patterns.

9.3 Why it can scale

- **Low marginal cost:** pose estimation runs on the user's device, so a new user costs little more than storing a few numbers.
- **Reach:** a web link or QR code is enough; there is no app store, and the app runs on older Android phones.
- **New languages are cheap:** each language is a translation file, and tests confirm every screen and voice cue is covered. Kiswahili, Hausa, Amharic, isiZulu, Arabic, French and Portuguese would open East, West, North and Southern Africa, alongside further Zimbabwean languages such as Tonga and Kalanga.
- **New exercises are configuration:** each exercise is a configuration object, so physiotherapists can request new ones without changes to the engine.
- **Portable hosting:** the backend can run on a university or in-country server to meet data-residency rules.

10. Implementation plan and milestones

The 12-month plan runs through 2027, the period of the first-place package. The first quarter makes the AI and the platform pilot-ready. A six-month supervised pilot follows, and the final quarter turns its results into a validation report and the first paid plans.

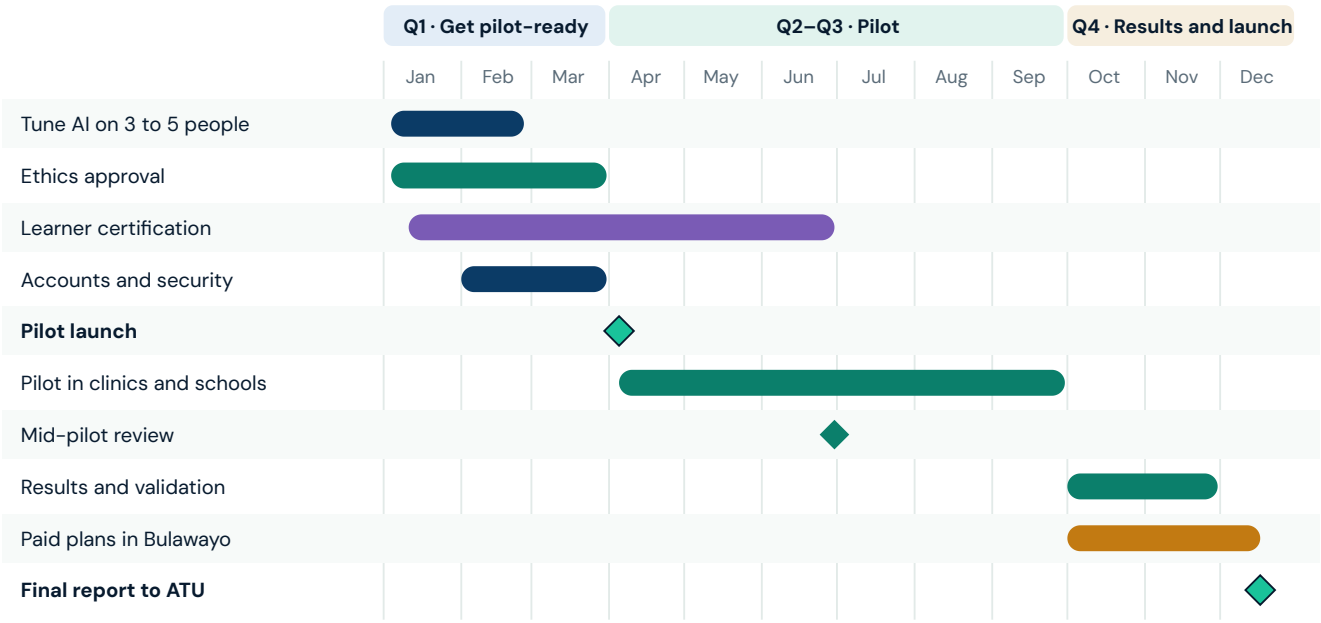


Figure 2. The pilot starts in April 2027 after tuning and ethics approval, and reports to ATU in December.

The pilot starts only after ethics approval and real-person tuning are complete. A shorter package would fund the same steps on a smaller pilot.

10.1 Milestones for staged support

Milestone	Due	Evidence for ATU
M1: Pilot-ready platform	End of March 2027	Tuned thresholds and their test recordings; ethics approval letter; accounts and a persistent database live
M2: Pilot running	End of June 2027	At least 3 sites active; enrolment against the targets of 50 patients and 60 athletes; mid-pilot review
M3: Pilot complete	End of September 2027	Anonymised session data; feedback from patients, physiotherapists and coaches
M4: Results and launch	December 2027	Validation report; first paying practices or schools; certified learners; final report to ATU

11. Budget and use of the support package

The first-place package of US$30,000 would fund the 12-month plan above. It follows the categories ATU's packages may cover. Figures are indicative and will be finalised in the costed support plan.

Category	What it pays for	US$
Cash facilitation to the innovator (30%, set by ATU)	The innovator's time and living costs while running the pilot	9,000
Equipment	10 Android phones and 2 iPhones for testing and for loan to patients without a suitable phone; 3 tablets with stands for clinic sites; 1 development laptop	6,000

Category	What it pays for	US$
Technical mentorship, cybersecurity and product development	Independent security and data-protection review; a physiotherapist mentor to review every exercise; native-speaker and accessibility user testing	6,000
Software, cloud and testing resources	Cloud hosting and database for 12 months; mobile data for pilot participants; participant transport and ethics review fees; developer tools	5,000
Training and certification	Certification for 4 learners in cloud, applied AI, cybersecurity and product management	4,000
Total		**30,000**

Why the equipment is needed. The AI must be proven on the phones patients actually own, from older Android models to iPhones. Loan phones let patients without a suitable phone join the pilot, so the results are not limited to wealthier users.

With a smaller package. A second-place package (US$20,000 over six months) would keep the tuning, security review and a pilot at 2 sites. A third-place package (US$10,000) would fund tuning, test devices and a 1-site pilot.

12. Expected impact and how it will be measured

The pilot will show whether RecoverLens gets more patients exercising correctly, with fewer trips, while their physiotherapist keeps watch. Most measures come from the app itself, so they cost nothing extra to collect.

Outcome	Measure	Pilot target
Patients keep exercising	Share of prescribed sessions completed	At least 70%
Exercises are done correctly	Share of reps with good form, week 1 against week 6	A clear rise for most patients
The AI agrees with the clinician	Rep counts against a physiotherapist's count on recorded sessions	At least 90% agreement
Pain is caught early	Pain rises after exercise reviewed by the physiotherapist	Within 2 working days
Travel is saved	Clinic trips replaced by remote review	Counted per patient
Physiotherapists reach more people	Patients followed between visits per physiotherapist	Compared with before the pilot
People find it easy	System Usability Scale score from patients and caregivers	70 or above
Athletes are screened	Athletes screened, and flagged patterns followed up by coaches	60 athletes screened
The ecosystem grows	Learners trained and certified	4 learners

12.1 Who benefits

- **Patients and families:** guided recovery at home, in their own language, without the cost of travel.
- **Physiotherapists:** visibility between visits and a way to focus on the patients who need them.
- **The health system:** more patients served by the same workforce, with data to plan rehabilitation services.
- **Code for Better and NUST learners:** a real health AI product to learn from, and a model for future student ventures.

13. Risks, ethics and data protection

The main risks are an AI that misreads a movement and a patient who over-exercises at home. Both are handled by keeping a physiotherapist in the loop and by testing on real people before the pilot.

Risk	Mitigation
The AI misreads a movement in poor light, baggy clothes or at an odd camera angle	The app checks the whole body is in view before a set; thresholds are tuned on real people; real recordings become regression tests; results are worded as observations for the physiotherapist
A patient over-exercises or is hurt at home	Only exercises a physiotherapist prescribes; a pain check-in before each session; alerts when pain rises; the coach suggests rest when reps show tiring
Personal health data is exposed	Video never leaves the device; only session numbers are stored; consent at sign-up; an independent security review; access by role
Patients lack a suitable phone or data	Runs on older phones and offline; loan phones during the pilot
Patients stop using it	Voice coaching, caregiver mode, WhatsApp sharing and physiotherapist follow-up on missed sessions
Regulation of software used in care	Early engagement with the Medicines Control Authority of Zimbabwe and the council that registers physiotherapists (MRPCZ); clinician-supervised use only
Translation errors	Review of all chiShona and isiNdebele text by native speakers before the pilot
Reliance on one developer	Learners from Code for Better join the team; the code has full technical documentation and automated tests

13.1 Ethics

The pilot protocol will be submitted to the NUST research ethics committee and the Medical Research Council of Zimbabwe before any patient is enrolled. Participants will give informed consent in English, chiShona or isiNdebele, and can withdraw at any time.

13.2 Data protection

RecoverLens is designed to comply with Zimbabwe's Cyber and Data Protection Act. It collects the minimum data needed: no video, no images and no body landmarks leave the device. Data is encrypted in transit, and the backend can be hosted in Zimbabwe.

14. Conclusion and the ask

RecoverLens shows that a Zimbabwean university ecosystem can build health AI that is useful, private and made for African patients. The platform already works. What it needs now is evidence from real clinics and homes, and the partners to carry it beyond Bulawayo.

Code for Better and RecoverLens ask ATU for:

1. **A place in the continental final** in Zimbabwe on 7–8 December 2026, to demonstrate RecoverLens live.
2. **The support package** to fund the 12-month pilot, validation and learner certification set out above.
3. **Introductions through ATU's networks** to mobile network operators, to explore zero-rated data for rehabilitation sessions. Introductions to regulators and to partners in other Member States would prepare the regional rollout.
4. **Training and certification places** for the learners who will carry RecoverLens forward.

> **Try it:** recover-lens.vercel.app

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